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# Bus Ticket Booking System (C++)

## Project title

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Note: Slides (excluding Thank-you and Introduction slide) should not be more than 4.

# Problem Statement

Managing bus seat reservations manually can lead to errors such as double booking, incorrect seat allocation, and difficulty in tracking available seats.

Traditional systems require a simple automated solution to efficiently manage seat reservations and cancellations.

This project aims to design a Bus Ticket Booking System using C++ that allows users to :-

- View available seats
- Book a seat
- Cancel a booked seat
- Update seat availability dynamically

The system works as a menu-driven console application that manages seat status during program execution.

# Proposed Solution

The solution is a **C++ based Bus Ticket Booking System** that simulates a real bus reservation process.

Key features of the system :-

- Stores bus number and total seats
- Displays seat layout with available and booked seats
- Allows users to book seats
- Allows users to cancel bookings
- Prevents invalid seat numbers and duplicate bookings

Seat status is stored using a **vector data structure**, where :-

- represents Available seat
- represents Booked seat

The system updates seat availability dynamically as users perform operations.

# Working (Method / Mind Map)

System Working :-

1. User enters: :- Bus Number Total Seats
2. System creates a **Bus object** and initializes seat status.
3. Menu options are displayed :-

    - View Seats
    - Book Seat
    - Cancel Seat
    - Exit

4. Operations performed :-

View Seats → Displays seat layout (2 seats + aisle + 2 seats)

Book Seat → Checks availability and marks seat as booked

Cancel Seat → Changes booked seat back to available

5. Seat status is updated using :-

- vector<bool> seats

The program continues until the user chooses **Exit**.

# Conclusion

The Bus Ticket Booking System demonstrates how a simple reservation system can be implemented using C++.

The system successfully manages seat allocation using a menu-driven interface and dynamic seat tracking.

Key programming concepts used include :-

- Object-Oriented Programming (Class Bus)
- Data structures (vector)
- Conditional statements for validation
- Loops for continuous user interaction

This project provides a basic understanding of how real-world transportation booking systems manage reservations efficiently.

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# Thank You